

The Prognostic Value of Serum Interleukin-6 Concentrations for 9 Cardiovascular and Mortality Outcomes

The Cross Cohort Collaboration (CCC)

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ABSTRACT

BACKGROUND Interleukin (IL)-6 has emerged as a promising target for cardiovascular disease (CVD) prevention. Better understanding IL-6's contribution to CVD events in community-based, diverse cohorts and in clinically relevant subgroups will provide critical context about IL-6 inhibition for CVD prevention.

OBJECTIVES The aim of this study was to evaluate the association of circulating IL-6 concentrations with incident cardiovascular events and mortality using pooled data from large multiethnic prospective cohorts.

METHODS The authors harmonized individual-level participant data from 14 cohorts with blood IL-6 measurements and follow-up data on any of 9 prespecified outcomes: myocardial infarction, stroke, atrial fibrillation, heart failure, total coronary heart disease (CHD), total CVD, all-cause mortality, CVD-specific mortality, and CHD-specific mortality. Multivariable-adjusted Cox proportional hazards models were used to estimate HRs and corresponding 95% CIs. IL-6 was analyzed by quartiles and as a log-transformed continuous variable. Associations were further examined by chronic kidney disease status, diabetes status, high-sensitivity C-reactive protein (hsCRP) concentrations, body mass index categories, as well as in secondary prevention. Discrimination was evaluated using the area under the curve across 4 models: base, base plus IL-6, base plus hsCRP, and base plus both markers.

RESULTS Among 59,396 participants, the median IL-6 concentration was 1.91 pg/mL. The mean age was 63.6 ± 12.4 years. 67.6% were women, and 20.6% were Black. The longest median follow-up time was 15.8 years (for CVD-specific mortality). Higher IL-6 concentrations were associated with increased risk for all 9 outcomes. The strongest association was observed for CHD-specific mortality, with an adjusted HR of 2.12 (95% CI: 1.88-2.39) in the highest compared with the lowest IL-6 quartile. The weakest association was for myocardial infarction (HR: 1.45; 95% CI: 1.28-1.64). Findings were robust and consistent in all key subgroups as well as in secondary prevention. IL-6 alone demonstrated modest additive discrimination beyond hsCRP for stroke, heart failure, CVD, CHD, and mortality.

CONCLUSIONS In this large, harmonized, individual-level participant data analysis of prospective cohorts, higher IL-6 concentrations were strongly associated with 9 cardiovascular and mortality outcomes after controlling for clinical covariates. These associations were consistent across cardiometabolic subgroups, chronic kidney disease status, and secondary prevention populations, highlighting the broad and consistent role of IL-6-mediated inflammation in cardiovascular risk. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS

AF = atrial fibrillation

AUC = area under the receiver-
operating characteristic curve

BMI = body mass index

CCC = Cross Cohort
Collaboration

CHD = coronary heart disease

CKD = chronic kidney disease

CRP = C-reactive protein

CVD = cardiovascular disease

HF = heart failure

hsCRP = high-sensitivity
C-reactive protein

IL = interleukin

IPD = individual-level
participant data

MI = myocardial infarction

Cardiovascular disease (CVD) remains the leading cause of morbidity and mortality worldwide, responsible for nearly one-third of all global deaths.¹ Although advances in the management of hypertension, dyslipidemia, smoking, and diabetes have substantially reduced cardiovascular risk,² many individuals continue to experience events, suggesting that additional factors, including systemic inflammation, may contribute.³

Among the array of inflammatory markers, interleukin (IL)-6 has emerged as an upstream cytokine with potential prognostic value and therapeutic relevance for cardiovascular risk reduction. Experimental studies have demonstrated that IL-6 is produced by vascular macrophages, fibroblasts, and endothelial cells in response to inflammatory stimuli such as NLRP3 inflammasome activation, IL-1 β , oxidative stress, and angiotensin II.⁴ In murine models, IL-6 promotes endothelial activation, thrombopoiesis, tissue factor expression, vascular permeability, and procoagulant activity, all of which contribute to atherogenesis and plaque destabilization.³ As a pleiotropic cytokine, IL-6 activates the Janus kinase/signal transducer and activator of transcription pathway, leading to hepatic synthesis of C-reactive protein (CRP), endothelial dysfunction, enhanced thrombogenicity, and plaque instability.⁵

Evidence from Mendelian randomization studies suggests that genetically mediated reductions in IL-6 signaling are associated with lower risk for coronary artery disease, stroke, atrial fibrillation (AF), and

all-cause mortality.⁶⁻⁸ Critically, clinical validation of inhibiting the NLRP3 inflammasome/IL-1 β /IL-6 pathway came from CANTOS (Canakinumab Anti-Inflammatory Thrombosis Outcomes Study), which demonstrated that IL-1 β inhibition with canakinumab significantly reduced major cardiovascular events in patients with prior myocardial infarction (MI) and elevated high-sensitivity CRP (hsCRP). Notably, the magnitude of IL-6 reduction predicted the degree of risk reduction, reinforcing IL-6 as a potential target for therapeutic modulation in CVD management and prevention.⁹ Building upon these findings, the RESCUE (Trial to Evaluate Reduction in Inflammation in Patients With Advanced Chronic Renal Disease Utilizing Antibody Mediated IL-6 Inhibition) trial showed that direct blocking of IL-6 produced dose-dependent reductions in hsCRP and other inflammatory biomarkers in patients with atherosclerosis and chronic kidney disease (CKD).¹⁰ Several pivotal trials are currently under way to evaluate IL-6-targeted therapies and to determine whether IL-6 inhibition leads to reductions in clinical cardiovascular event rates (NCT06118281, NCT05636176).¹¹

While awaiting the findings from those trials, further understanding of the contribution of IL-6 in CVD event rates in community-based, diverse cohorts of adults with average CVD risk, as well as those in clinically relevant subgroups, will provide critical context about IL-6 inhibition for CVD prevention. Prior epidemiologic studies have consistently shown that IL-6 is associated with increased risk for CVD events and mortality, yet uncertainties remain because single cohort results often are inadequate to address certain questions. Particularly understudied

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are subgroups defined by the presence of CKD, diabetes, severe obesity, or known elevations in baseline inflammation. These populations represent a growing segment of the adult population and bear a disproportionate burden of inflammation-related CVD. Harmonizing individual-level participant data (IPD) from multiple prospective epidemiologic cohorts is a technique that offers the ability to better characterize IL-6 associated CVD risk. Analysis of harmonized IPD is superior to meta-analysis of aggregated data through avoiding publication bias, avoiding heterogeneity of covariate adjustment, reducing sparse data biases, reducing misclassification of exposures and outcomes by using common definitions of each, and enabling analysis of subgroups and dose-response characteristics with greater statistical precision.¹²⁻¹⁴ Because high-quality harmonization of multiple large cohort data structures is laborious, requiring technical expertise and rigorous methodology, it has been used sparingly in cardiovascular epidemiology. Nevertheless, harmonized IPD analyses across multiple cohorts provide an opportunity to increase statistical power to evaluate clinically meaningful subgroups and establish population-level benchmarks that may help contextualize results from emerging trials targeting the IL-6 inflammatory pathway.

In the present study, we evaluated the association of IL-6 with incident cardiovascular events and mortality in a large, pooled data set of 14 community-based cohorts characterized by average baseline CVD risk. Our objective was to quantify the prognostic significance of IL-6 both in general and within clinically meaningful subgroups, assess whether IL-6 offers enhanced risk prediction compared with the more commonly used hsCRP and potentially provide evidence to help identify subpopulations likely to benefit from targeted IL-6 blockade interventions.

METHODS

This study was approved by the Johns Hopkins Medicine Institutional Review Board (IRB00474332) and conducted in accordance with the principles of the Declaration of Helsinki.

STUDY POPULATION. We conducted a prospective, observational analysis using harmonized IPD from the Cross Cohort Collaboration (CCC), a multicohort initiative comprising 25 predominantly U.S.-based prospective studies funded by the National Heart, Lung, and Blood Institute (NHLBI) and other National Institutes of Health institutes. The CCC infrastructure, previously used in investigations of tobacco use and obesity-related cardiometabolic risk, integrates

cohort-level IPD into a unified, quality-controlled data set through a rigorously standardized harmonization protocol¹⁵ consistent with conventional IPD harmonization procedures in epidemiologic investigations.¹³ Cohort characteristics have been outlined in the design paper.¹⁵

Following proposal approval, data-use agreements were established with each contributing cohort. All data were transferred to a secure, encrypted server at Johns Hopkins University, where harmonization and quality control were performed using a fully annotated and reproducible algorithm.

For this analysis, participants were eligible for inclusion if they had IL-6 measurements and follow-up data for at least 1 of 9 prespecified cardiovascular or mortality outcomes. The study population comprised 59,396 participants from 14 individual community-based cohorts, including the ARIC (Atherosclerosis Risk In Communities) study, the BLSA (Baltimore Longitudinal Study of Aging), the CHS (Cardiovascular Health Study), the CRIC (Chronic Renal Insufficiency Cohort) study, CARDIA (Coronary Artery Risk Development in Young Adults) study, the Framingham Offspring Study, Framingham Heart Study Third Generation, the Health ABC (Health, Aging, and Body Composition) study, MESA (Multi-Ethnic Study of Atherosclerosis), the MrOS (Osteoporotic Fractures in Men) study, RBS (Rancho Bernardo Study of Healthy Aging), the REGARDS (Reasons for Geographic And Racial Differences in Stroke) study, the WHI (Women's Health Initiative)-Observational Study, and the WHI Clinical Trial.

IL-6 MEASUREMENT. The primary exposure was baseline blood IL-6 concentrations, harmonized across cohorts and expressed in picograms per milliliter (pg/mL), as described in a previous CCC publication.¹⁶ Of note, "baseline" refers here to the harmonized baseline defined within the CCC data set (time zero, when follow-up begins) and may not correspond to the baseline visit of each individual cohort. All 14 cohorts measured IL-6 using high-sensitivity enzyme-linked immunosorbent assays from R&D Systems ([Supplemental Table 1](#)), with interassay coefficients of variation largely between 6% and 10% (ranging up to ~13%) and intra-assay variability typically <7%, supporting overall assay reliability. For the main analyses, IL-6 was evaluated both as a continuous variable (log transformed to normalize its distribution) and as a categorical variable in quartiles.

OUTCOMES ASCERTAINMENT. We evaluated an a priori defined broad spectrum of adjudicated clinical cardiovascular endpoints and mortality outcomes,

harmonized as consistently as possible across cohorts. Outcome definitions were standardized using prespecified criteria established in prior CCC publications.¹⁷

Harmonized cardiovascular outcomes were grouped into individual and composite categories. Individual events included MI, stroke, heart failure (HF), and AF. Composite outcomes included incident total CVD and incident total coronary heart disease (CHD). The composite of CVD was defined as all atherosclerotic CVD events, including CHD, stroke, or cardiovascular mortality (coronary mortality, stroke mortality, other atherosclerotic mortality, or other CVD mortality). The composite of CHD was defined as the total of coronary revascularization, MI, or coronary mortality. Not every cohort contributed data for all 9 outcomes, as shown in [Supplemental Table 2](#).

Mortality outcomes comprised all-cause mortality, CVD-specific mortality, and CHD-specific mortality. In outcome analysis, individuals with prevalent disease for the specific outcome of interest at baseline were excluded to ensure the accurate ascertainment of incident events.

OTHER COVARIATES. Covariate selection was guided by established associations with cardiovascular and mortality risks. All variables were harmonized following best practices previously detailed in CCC publications,¹⁵⁻¹⁷ and consistent with conventional IPD harmonization and decision methodology in epidemiologic studies.¹³ Demographic covariates included age, sex, and race/ethnicity (categorized as “White,” “Black,” “Hispanic,” “Asian,” or “American or Alaskan Indian”). Educational attainment was categorized as “less than high school,” “completed high school,” “some college, graduate, or advanced degree,” and “other,” indicating missing data. Clinical covariates included body mass index (BMI); systolic blood pressure; diastolic blood pressure; prevalent conditions including hypertension, diabetes, and hyperlipidemia; use of antihypertensive and lipid-lowering medications; total cholesterol level; low-density lipoprotein cholesterol level; and baseline CHD status. Lifestyle factors included smoking status (categorized as “never,” “former,” and “current”), and alcohol use (yes vs no). hsCRP concentrations were harmonized in units of milligrams per liter (mg/L) and included as a commonly used measure of systemic inflammation.¹⁶ CKD was defined as a composite of estimated glomerular filtration rate 15 to <60 mL/min/1.73 m² (CKD-EPI 2021 creatinine equation) or estimated glomerular filtration rate ≥60 mL/min/1.73 m² with urinary albumin-to-creatinine ratio ≥200 mg/g. Variables

with >10% missingness were imputed within each cohort for descriptive analyses only, using regression-based models informed by relevant demographic and clinical correlates.

STATISTICAL ANALYSIS. Baseline characteristics were described by IL-6 quartile using mean ± SD, median (Q1-Q3), or proportions, as appropriate.

Crude incidence rates for each outcome were calculated overall and across IL-6 quartiles. Time-to-event associations between IL-6 and outcomes were assessed using Cox proportional hazards models with sequential multivariable adjustment. Assessment of Schoenfeld residuals indicated that the Cox proportional hazards assumption was met. Covariates were introduced sequentially to reflect a conceptual hierarchy, beginning with demographic characteristics, followed by established cardiovascular risk factors and comorbidities, as well as hsCRP. Model 1 included age, sex, race/ethnicity, and education level. Model 2 further adjusted for BMI, systolic and diastolic blood pressure, prevalent hypertension, use of antihypertensive medications, prevalent diabetes, prevalent hyperlipidemia, use of lipid-lowering medications, smoking status, alcohol use, and baseline CHD status. Model 3 additionally adjusted for hsCRP concentrations. Participants were followed from time zero until the time of the event, death, loss to follow-up, or the end of the follow-up period for each cohort, whichever came first. HRs and corresponding 95% CIs were calculated. Pearson correlation coefficients were calculated to assess the linear relationship between log_e(IL-6) and log_e(hsCRP) concentrations. We also constructed multivariable Cox proportional hazards models to evaluate IL-6 and hsCRP per SD increase on the log-transformed scale, enabling a more direct comparison of their associations with CVD and mortality outcomes. Models were adjusted for the complete set of covariates, including age, sex, race/ethnicity, education level, BMI, systolic blood pressure, diastolic blood pressure, prevalent hypertension, antihypertensive medication use, prevalent diabetes, prevalent hyperlipidemia, lipid-lowering medication use, smoking status, alcohol use, and baseline CHD status. To account for unobserved cohort-level heterogeneity and intra-cohort correlation in event times, Cox proportional hazards models were fitted with a shared frailty term at the cohort level. The frailty term represents a latent cohort-specific random effect, allowing participants within the same cohort to share a common baseline risk. Frailty was specified using a gamma distribution, and HRs were estimated conditional on the cohort-level frailty.

We additionally evaluated the dose-response characteristics between IL-6 concentrations and 9 outcomes by depicting restricted cubic spline curves. Five knots were selected at 0.60, 1.18, 1.91, 3.18, and 7.18 pg/mL, corresponding to the 5th, 25th, 50th, 75th, and 95th percentiles, respectively, with IL-6 values winsorized at 7.20 pg/mL to minimize the influence of extreme outliers. Restricted cubic spline models were adjusted for age, sex, race/ethnicity, educational level, and individual cohort. We used a parsimonious adjustment set in restricted cubic spline models to preserve stability and avoid over-adjustment, while reserving full covariate adjustment for primary models to evaluate robustness.

Time-to-event data were administratively censored at 10 years, and a binary outcome indicating event occurrence within this time horizon was defined. Logistic regression models were used to estimate predicted 10-year risk, adjusting for age, sex, race/ethnicity, and cohort. Sequential models additionally incorporated: 1) log-transformed hsCRP; 2) log-transformed IL-6; and 3) log-transformed hsCRP plus log-transformed IL-6. Model discrimination was assessed using the area under the receiver-operating characteristic curve (AUC), and differences between models were compared using the DeLong test for correlated receiver-operating characteristic curves. We then evaluated the incremental predictive value of IL-6 beyond the base model using categorical net reclassification improvement across 9 outcomes. Predicted 10-year stroke risk was estimated for each participant under 2 models: the base model and the base model plus IL-6. Participants were stratified into 3 risk groups using clinically meaningful thresholds appropriate to this study population: <5%, 5% to <10%, and ≥10% for MI, stroke, AF, HF, CHD, CVD-specific mortality, and CHD-specific mortality; <10%, 10% to <20%, and ≥20% for CVD; and <15%, 15% to <30%, and ≥30% for all-cause mortality. Cutpoints were selected to ensure adequate numbers of events within each category for stable estimation in this pooled study cohort; they were not intended to represent guideline-based treatment thresholds. We then cross-tabulated the distribution of cases across categories between the 2 models. The percentage of cases reclassified, as well as the overall net reclassification improvement, were calculated to quantify improvement in risk discrimination according to IL-6 level.

Prespecified subgroup analyses were conducted across strata defined by CKD status (with CKD vs without CKD), diabetes status (nondiabetic, prediabetic, and diabetic), systemic inflammation status based on hsCRP concentrations (<2 mg/L vs ≥2 mg/L

and <3 mg/L vs ≥3 mg/L), and BMI categories (normal weight [18.5-24.9 kg/m²], overweight [25.0-29.9 kg/m²], class I obesity [30.0-34.9 kg/m²], class II obesity [35.0-39.9 kg/m²], and class III obesity [≥40.0 kg/m²]), and age groups (<65 years vs ≥65 years). Dedicated analyses were also conducted among participants with baseline CVD to evaluate associations in a secondary prevention population. Effect modification between subgroup variables and IL-6 concentrations was evaluated using multiplicative interaction terms across all 9 outcomes. Stratified HRs with 95% CIs were calculated to show the associations across subgroups.

To ensure robustness, we performed sensitivity analyses using 1) individual cohort-specific IL-6 quartiles instead of pooled cohort quartiles; and 2) standardized IL-6 concentrations recentered within each cohort. For the latter, IL-6 values were adjusted within strata defined by sex and age (<60 years vs ≥60 years). In each stratum, we calculated the overall median across the study population and the cohort-specific median, then derived a recentering coefficient as the ratio of the overall to the cohort-specific median. All IL-6 values in that stratum were multiplied by the corresponding coefficient, thereby aligning cohort-specific distributions to the overall study population. We additionally conducted a sensitivity analysis excluding the REGARDS cohort, as IL-6 was measured in a subsample, and weighting by sampling fractions to represent the full cohort was not applied in the primary analysis. We also performed a sensitivity analysis excluding events within the first 2 years after baseline to minimize potential reverse causation. We additionally conducted a sensitivity analysis to obtain population-level estimates using standard Cox models that included cohort as a fixed-effect covariate with robust SEs clustered by cohort, rather than modeling cohort as a random effect through shared frailty.

All statistical tests were considered significant at a 2-sided *P* value <0.05. To avoid inflation of type I error due to multiple comparisons, Bonferroni-adjusted *P* values were checked and are reported where applicable, except for interaction analyses. Analyses were conducted using Stata version 18 (StataCorp).

RESULTS

BASELINE CHARACTERISTICS OF THE STUDY POPULATION. As shown in Table 1, this study included 59,396 participants with a mean age of 63.6 ± 12.4 years, of whom 37,923 (67.6%) were

TABLE 1 Baseline Characteristics of the Study Population by Quartiles of IL-6

	Quartiles of IL-6				Total (N = 59,396)
	First (n = 14,859 [25.0%])	Second (n = 14,848 [25.0%])	Third (n = 14,822 [25.0%])	Fourth (n = 14,867 [25.0%])	
IL-6, pg/mL	0.85 (0.65-1.01)	1.52 (1.34-1.71)	2.43 (2.14-2.76)	4.66 (3.76-6.51)	1.91 (1.18-3.18)
Age, y	59.1 ± 12.8	63.2 ± 12.2	65.4 ± 11.7	66.6 ± 11.7	63.6 ± 12.4
Sex					
Female	10,203 (69.7)	9,638 (68.1)	9,263 (67.0)	8,819 (65.2)	37,923 (67.6)
Male	4,432 (30.3)	4,508 (31.9)	4,559 (33.0)	4,710 (34.8)	18,209 (32.4)
Race/ethnicity					
White	10,717 (72.5)	10,947 (74.0)	10,665 (72.2)	10,290 (69.5)	42,619 (72.1)
Black	2,337 (15.8)	2,753 (18.6)	3,236 (21.9)	3,850 (26.0)	12,176 (20.6)
Asian	896 (6.1)	322 (2.2)	194 (1.3)	128 (0.9)	1,540 (2.6)
Hispanic	667 (4.5)	600 (4.1)	474 (3.2)	355 (2.4)	2,096 (3.5)
American or Alaskan Indian	165 (1.1)	170 (1.1)	193 (1.3)	173 (1.2)	701 (1.2)
Education level					
Less than high school	1,216 (8.2)	1,608 (10.8)	1,685 (11.4)	1,695 (11.4)	6,204 (10.4)
Completed high school	2,530 (17.0)	3,002 (20.2)	3,242 (21.9)	3,403 (22.9)	12,177 (20.5)
Some college, graduate, or advanced degree	10,479 (70.5)	9,274 (62.5)	8,560 (57.8)	8,160 (54.9)	36,473 (61.4)
Other	634 (4.3)	964 (6.5)	1,335 (9.0)	1,609 (10.8)	4,542 (7.6)
Body mass index, kg/m ²	25.7 ± 4.5	27.5 ± 5.1	28.9 ± 6.0	30.3 ± 7.0	28.1 ± 6.0
Systolic blood pressure, mm Hg	124.1 ± 18.8	129.0 ± 20.0	131.5 ± 20.4	132.0 ± 20.6	129.0 ± 20.2
Diastolic blood pressure, mm Hg	72.9 ± 10.0	73.5 ± 10.4	74.0 ± 10.8	73.6 ± 11.0	73.5 ± 10.5
Prevalent hypertension	5,046 (34.0)	6,908 (46.6)	8,172 (55.2)	9,024 (60.9)	29,150 (49.1)
Antihypertensive medication use	3,249 (22.0)	4,750 (32.2)	5,909 (40.2)	6,964 (47.4)	20,872 (35.4)
Prevalent diabetes	1,122 (7.6)	1,942 (13.1)	2,796 (18.9)	3,695 (24.9)	9,555 (16.1)
Antidiabetic medication use	589 (4.5)	920 (7.5)	1,242 (11.0)	1,522 (14.5)	4,273 (9.1)
Serum triglyceride, mg/dL	98 (70-141)	111 (79-160)	115 (83-162)	116 (84-164)	110 (79-158)
Total cholesterol, mg/dL	197.2 ± 37.7	198.4 ± 39.7	195.0 ± 40.6	188.8 ± 42.0	194.7 ± 40.3
LDL cholesterol, mg/dL	117.2 ± 33.1	118.3 ± 34.4	115.8 ± 35.3	111.4 ± 36.4	115.6 ± 35.0
HDL cholesterol, mg/dL	56.6 ± 16.7	53.7 ± 16.1	52.1 ± 15.8	50.4 ± 15.4	53.1 ± 16.2
Prevalent hyperlipidemia	2,656 (18.9)	3,193 (22.9)	3,275 (23.1)	3,186 (22.2)	12,310 (21.8)
Lipid-lowering medication use	1,882 (12.8)	2,546 (17.3)	3,205 (22.1)	3,873 (27.1)	11,506 (19.8)
Alcohol use	8,741 (59.2)	7,800 (52.8)	7,219 (49.0)	6,718 (45.5)	30,478 (51.6)
Smoking status					
Never	8,030 (54.4)	7,221 (49.1)	6,616 (45.4)	6,269 (43.1)	28,136 (48.0)
Former	5,417 (36.7)	5,895 (40.1)	6,213 (42.6)	6,302 (43.3)	23,827 (40.7)
Current	1,314 (8.9)	1,581 (10.8)	1,758 (12.1)	1,984 (13.6)	6,637 (11.3)
History of coronary heart disease	457 (3.2)	854 (6.5)	1,025 (8.6)	1,161 (10.6)	3,497 (6.9)
History of stroke	153 (1.0)	265 (1.8)	349 (2.4)	390 (2.6)	1,157 (1.9)
History of cardiovascular disease	578 (3.6)	1,031 (6.6)	1,288 (8.3)	1,425 (9.3)	4,322 (6.9)
History of heart failure	73 (0.5)	183 (1.2)	285 (1.9)	381 (2.6)	922 (1.6)
History of chronic kidney disease	1,417 (14.8)	2,051 (21.5)	2,439 (25.5)	2,890 (30.5)	8,797 (23.1)
hsCRP level, mg/L	1.72 ± 2.27	2.74 ± 3.45	3.83 ± 4.47	7.25 ± 9.95	3.91 ± 6.23

Values are median (Q1-Q3), mean ± SD, or n (%).

HDL = high-density lipoprotein; hsCRP = high-sensitivity C-reactive protein; IL = interleukin; LDL = low-density lipoprotein.

women and 12,176 (20.6%) were Black. The median IL-6 concentration was 1.91 pg/mL (Q1-Q3: 1.18-3.18 pg/mL). Participants with higher IL-6 concentrations were older and exhibited a higher prevalence of hypertension, diabetes, CVD, and CKD. They were also more likely to use antihypertensive and

lipid-lowering medications, had higher BMI, and were more frequently current smokers or alcohol users. Concentrations of hsCRP increased with higher IL-6 quartiles. IL-6 and hsCRP were correlated, with a Pearson correlation coefficient of 0.441 (95% CI: 0.434-0.448). Pearson correlation coefficients

TABLE 2 Crude Incidence Rates (per 1,000 Person-Years) of Cardiovascular and Mortality Outcomes Overall and Across IL-6 Quartiles

	Overall	Quartiles of IL-6			
		Q1	Q2	Q3	Q4
MI	6.1 (5.9-6.3)	3.5 (3.2-3.8)	6.0 (5.7-6.8)	7.1 (6.7-7.6)	8.9 (8.4-9.4)
Stroke	6.7 (6.5-6.8)	3.5 (3.2-3.7)	6.2 (5.8-6.4)	8.2 (7.8-8.6)	10.2 (9.7-10.7)
AF	10.4 (10.2-10.7)	8.8 (8.3-9.4)	10.9 (10.3-11.6)	10.5 (9.9-11.1)	11.4 (10.8-12.1)
HF	7.6 (7.4-7.8)	3.7 (3.4-3.9)	6.3 (6.0-6.7)	9.3 (8.8-9.8)	13.5 (12.9-14.2)
CVD	15.1 (14.8-15.4)	8.5 (8.1-8.9)	14.7 (14.1-15.3)	19.1 (18.4-19.8)	23.9 (23.0-24.8)
CHD	8.9 (8.7-9.1)	5.1 (4.8-5.4)	8.8 (8.4-9.3)	10.9 (10.4-11.4)	13.7 (13.0-14.4)
All-cause mortality	32.8 (32.4-33.2)	20.4 (19.8-21.0)	30.0 (29.2-30.7)	37.8 (37.0-38.7)	47.5 (46.5-48.6)
CVD-specific mortality	8.8 (8.5-9.0)	4.3 (4.1-4.6)	7.9 (7.5-8.2)	10.7 (10.2-11.2)	14.7 (14.1-15.3)
CHD-specific mortality	5.3 (5.2-5.5)	2.7 (2.5-2.9)	5.0 (4.7-5.3)	6.6 (6.2-7.0)	8.7 (8.2-9.2)

Values are incidence rate (95% CI).

AF = atrial fibrillation; CHD = coronary heart disease; CVD = cardiovascular disease; HF = heart failure; IL = interleukin; MI = myocardial infarction.

stratified by individual cohorts are presented in [Supplemental Table 3](#).

INCIDENCE RATES OF CARDIOVASCULAR AND MORTALITY OUTCOMES. The median follow-up time varied across the 9 outcomes, ranging from 12.0 years (Q1-Q3: 8.0-16.8 years) for stroke to 15.8 years (Q1-Q3: 11.4-20.0 years) for CVD-specific mortality ([Supplemental Table 4](#)). Overall, the highest incidence rates were observed for all-cause mortality (32.8 per 1,000 person-years; 95% CI: 32.4-33.2) and second highest for the CVD composite (15.1 per 1,000 person-years; 95% CI: 14.8-15.4), while the lowest were for CHD-specific mortality (5.3 per 1,000 person-years; 95% CI: 5.2-5.5) and MI (6.1 per 1,000 person-years; 95% CI: 5.9-6.3) ([Table 2](#)). Incidence rates for cardiovascular and mortality outcomes increased progressively across IL-6 quartiles. For instance, the incidence of MI rose from 3.5 (95% CI: 3.2-3.8) to 8.9 (95% CI: 8.4-9.4) per 1,000 person-years from the Q1 to Q4 IL-6 quartile. A dose-response gradient was observed for all other outcomes.

ASSOCIATIONS BETWEEN IL-6 AND CARDIOVASCULAR OUTCOMES. As shown in [Table 3](#), higher IL-6 concentrations were consistently and independently associated with increased risk for cardiovascular and mortality outcomes across all models. These associations were attenuated from model 1 to model 3. When modeled as a continuous variable ($\log_e[\text{IL-6}]$), IL-6 showed the strongest association with incident HF among all cardiovascular outcomes in model 3, with an adjusted HR of 1.31 (95% CI: 1.24-1.38). Model 3 showed HRs of 1.28 (95% CI: 1.24-1.33) for composite CVD, 1.27 (95% CI: 1.22-1.33) for composite CHD, 1.30 (95% CI: 1.27-1.33) for all-cause mortality, 1.42 (95% CI: 1.36-1.48) for CVD-specific mortality,

and 1.46 (95% CI: 1.38-1.54) for CHD-specific mortality. When IL-6 was examined categorically, individuals in Q4 had significantly elevated risks compared with those in Q1 across all models. In model 3, the strongest association among individual cardiovascular outcomes remained HF, with an HR of 1.64 (95% CI: 1.46-1.85) for Q4 vs Q1. Compared with Q1, Q4 was associated with a 68% higher risk for composite CVD, 62% higher risk for composite CHD, and elevated risks of 70%, 104%, and 112% for all-cause, CVD-specific, and CHD-specific mortality, respectively. Restricted cubic spline curves demonstrated a generally dose-responsive relationship between IL-6 concentrations and all 9 outcomes ([Figure 1](#)).

Among the 4,277 participants with baseline CVD, the mean age was 70.0 ± 8.7 years, and the median IL-6 concentration was 2.38 pg/mL (Q1-Q3: 1.54-3.77 pg/mL) ([Supplemental Table 5](#)). In this secondary prevention population, higher IL-6 concentrations largely remained significantly associated with increased risk for incident cardiovascular events and mortality. For example, when modeled as a continuous variable ($\log_e[\text{IL-6}]$), fully adjusted HRs were 1.41 (95% CI: 1.27-1.55) for HF, 1.39 (95% CI: 1.31-1.48) for all-cause mortality, and 1.41 (95% CI: 1.26-1.57) for CHD-specific mortality, with no significant association observed for stroke. When IL-6 was modeled by quartiles, participants in Q4 consistently exhibited elevated risks compared with those in Q1 for all outcomes except stroke and MI ([Supplemental Table 6](#)).

PREDICTIVE VALUES AND COMPARISONS BETWEEN IL-6 AND hsCRP. Compared with hsCRP, IL-6 demonstrated a stronger and more consistent association across all outcomes ([Table 4](#)). Each SD increase in log-transformed IL-6 was associated with

TABLE 3 Risk for Cardiovascular and Mortality Outcomes by Baseline IL-6

Outcome (Incident Cases/Total Observations)	log _e (IL-6)	Quartiles of IL-6			
		Q1	Q2	Q3	Q4
MI (4,211/54,141)	Model 1	1.37 (1.31-1.43)	Reference	1.41 (1.28-1.55)	1.60 (1.46-1.76)
	Model 2	1.27 (1.21-1.34)	Reference	1.29 (1.17-1.43)	1.41 (1.27-1.56)
	Model 3	1.20 (1.14-1.27)	Reference	1.26 (1.14-1.40)	1.31 (1.17-1.46)
Stroke (4,747/56,721)	Model 1	1.34 (1.28-1.40)	Reference	1.39 (1.26-1.52)	1.65 (1.50-1.81)
	Model 2	1.25 (1.19-1.31)	Reference	1.29 (1.17-1.42)	1.44 (1.30-1.59)
	Model 3	1.20 (1.13-1.27)	Reference	1.26 (1.14-1.40)	1.36 (1.22-1.51)
AF (4,770/34,363)	Model 1	1.41 (1.36-1.47)	Reference	1.31 (1.20-1.42)	1.55 (1.42-1.69)
	Model 2	1.27 (1.21-1.34)	Reference	1.19 (1.08-1.31)	1.28 (1.15-1.41)
	Model 3	1.28 (1.21-1.35)	Reference	1.19 (1.08-1.31)	1.27 (1.15-1.42)
HF (5,204/52,804)	Model 1	1.58 (1.53-1.64)	Reference	1.32 (1.20-1.45)	1.83 (1.67-2.01)
	Model 2	1.35 (1.29-1.42)	Reference	1.16 (1.05-1.28)	1.39 (1.25-1.54)
	Model 3	1.31 (1.24-1.38)	Reference	1.15 (1.04-1.27)	1.33 (1.20-1.48)
CVD (9,375/42,315)	Model 1	1.40 (1.36-1.44)	Reference	1.36 (1.28-1.45)	1.63 (1.53-1.74)
	Model 2	1.31 (1.27-1.35)	Reference	1.26 (1.18-1.34)	1.43 (1.34-1.53)
	Model 3	1.28 (1.24-1.33)	Reference	1.26 (1.18-1.35)	1.40 (1.30-1.50)
CHD (5,713/43,089)	Model 1	1.42 (1.37-1.47)	Reference	1.39 (1.29-1.51)	1.62 (1.50-1.76)
	Model 2	1.32 (1.27-1.37)	Reference	1.27 (1.17-1.38)	1.41 (1.30-1.53)
	Model 3	1.27 (1.22-1.33)	Reference	1.25 (1.15-1.36)	1.35 (1.24-1.48)
All-cause mortality (26,701/58,950)	Model 1	1.39 (1.37-1.42)	Reference	1.22 (1.18-1.27)	1.49 (1.43-1.55)
	Model 2	1.30 (1.27-1.32)	Reference	1.18 (1.13-1.23)	1.37 (1.32-1.43)
	Model 3	1.30 (1.27-1.33)	Reference	1.19 (1.14-1.24)	1.38 (1.32-1.44)
CVD-specific mortality (6,531/48,626)	Model 1	1.58 (1.53-1.63)	Reference	1.43 (1.32-1.55)	1.82 (1.68-1.97)
	Model 2	1.44 (1.38-1.49)	Reference	1.31 (1.21-1.43)	1.55 (1.42-1.69)
	Model 3	1.42 (1.36-1.48)	Reference	1.32 (1.21-1.44)	1.53 (1.40-1.68)
CHD-specific mortality (3,986/48,626)	Model 1	1.63 (1.56-1.70)	Reference	1.48 (1.34-1.64)	1.92 (1.74-2.13)
	Model 2	1.49 (1.42-1.56)	Reference	1.36 (1.22-1.51)	1.63 (1.46-1.81)
	Model 3	1.46 (1.38-1.54)	Reference	1.34 (1.20-1.50)	1.58 (1.41-1.77)

Values are HR (95% CI). Model 1 was adjusted for sex, age, race/ethnicity, and education level. Model 2 was further adjusted for body mass index, prevalent hypertension, antihypertensive medication use, prevalent diabetes, prevalent hyperlipidemia, lipid-lowering medication use, systolic blood pressure, diastolic blood pressure, smoking status, alcohol use, and baseline CHD status. Baseline CHD status was not adjusted for the incident CHD outcome analysis. Model 3 was further adjusted for log_e(hsCRP). All reported associations were statistically significant ($P < 0.05$) and remained so after Bonferroni correction for multiple comparisons. Between-cohort variability was addressed by including a cohort-specific frailty term (random effect) in the survival models.

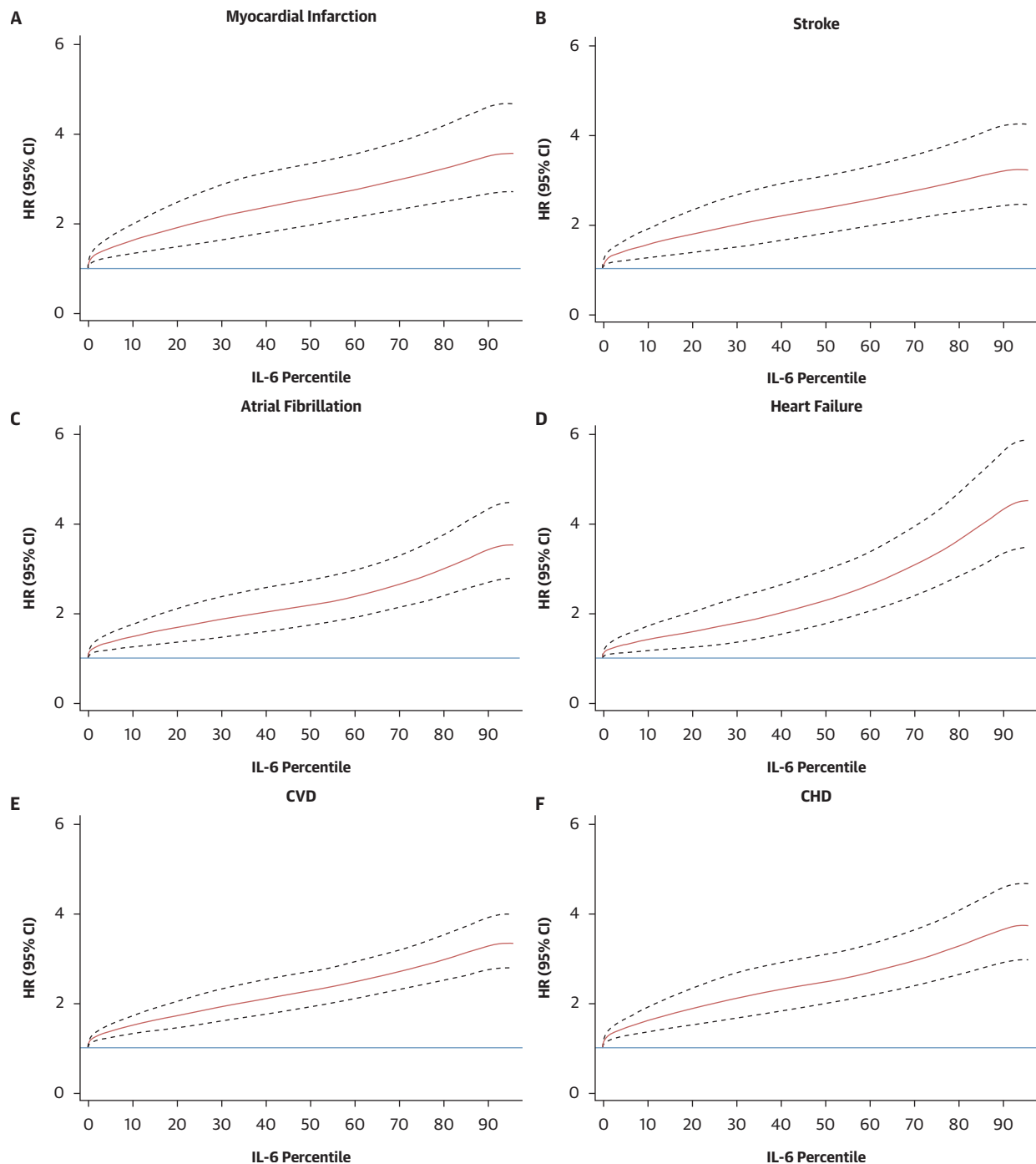
Abbreviations as in [Table 2](#).

higher risk for cardiovascular and mortality outcomes, whereas the associations for hsCRP were generally weaker and, in some cases, not statistically significant. For example, a 1-SD increase in log_e(IL-6) was associated with a 23% higher risk for HF (HR: 1.23; 95% CI: 1.18-1.28), while the corresponding increase in log_e(hsCRP) was associated with a 6% higher risk (HR: 1.06; 95% CI: 1.02-1.10). Notably, IL-6, but not hsCRP, was independently associated with incident AF and CHD-specific mortality.

In analyses of risk discrimination, the addition of IL-6 to the base model modestly improved prediction across all cardiovascular and mortality outcomes ([Table 5](#)). For instance, the AUC for HF increased from 0.732 (95% CI: 0.724-0.740) to 0.756 (95% CI: 0.748-0.763). Across outcomes, IL-6 generally conferred greater improvements in discrimination than hsCRP, although the magnitude of improvement was

relatively modest; adding both inflammation markers yielded small gains beyond IL-6 alone, except for MI. Differences in AUC compared with the base model were relatively small but statistically significant for all outcomes ($P < 0.05$), except AF in the model including IL-6. Consistent with these findings, incorporating IL-6 led to overall modest improvements in net reclassification across most outcomes ([Supplemental Tables 7 to 15](#)); however, no significant improvement was observed for AF.

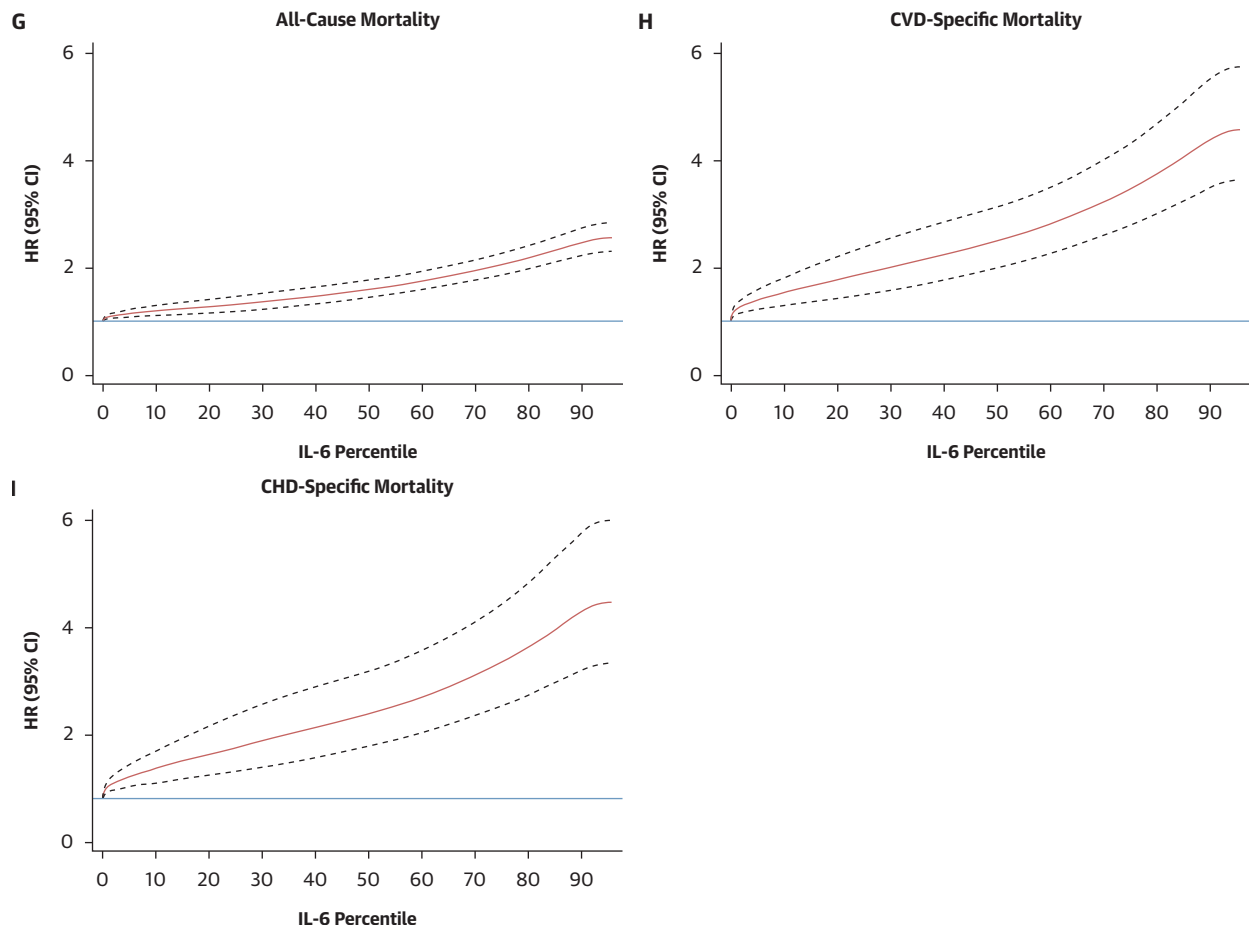
SUBGROUP ANALYSES. As shown in [Figure 2A](#), IL-6 was significantly associated with increased risk for all outcomes in both individuals with and without CKD, with consistently stronger point estimates observed among those with CKD. Statistically significant positive interactions between IL-6 and CKD status were noted for all-cause mortality and CVD-specific mortality. [Figure 2B](#) illustrates elevated risks across diabetes status strata. A significant

FIGURE 1 Restricted Cubic Splines of the Relationships Between IL-6 and Risk for Cardiovascular and Mortality Outcomes

(A-I) Restricted cubic spline models of interleukin (IL)-6 IL-6 percentile in relation to specific outcomes: myocardial infarction (A), stroke (B), atrial fibrillation (C), heart failure (D), composite cardiovascular disease (CVD) (E), composite coronary heart disease (CHD) (F), all-cause mortality (G), CVD-specific mortality (H), and CHD-specific mortality (I). The red solid line represents the multivariable-adjusted HR, the black dashed lines indicate the 95% CI, and the blue horizontal line marks the reference HR of 1.0. Five knots were selected at 0.60, 1.18, 1.91, 3.18, and 7.18 pg/mL, corresponding to the 5th, 25th, 50th, 75th, and 95th percentiles, respectively. IL-6 concentrations were Winsorized at 7.20 pg/mL. Analyses were adjusted for age, sex, race/ethnicity, education level, and individual cohorts.

Continued on the next page

FIGURE 1 Continued



positive interaction between IL-6 and diabetes status was observed for AF, HF, and all-cause mortality. In [Figure 2C](#), IL-6 was consistently associated with increased risk for cardiovascular and mortality outcomes across hsCRP strata. These associations were largely comparable between groups with hsCRP <2 mg/L vs $\geq$ 2 mg/L. Similar findings were observed when hsCRP was categorized as <3 mg/L vs $\geq$ 3 mg/L ([Supplemental Figure 1](#)). In BMI-stratified analyses, the associations between IL-6 and HF, CVD, CHD, and mortality outcomes were stable and significant across all BMI categories ([Figure 2D](#)). In contrast, for MI, stroke, and AF, IL-6 was significantly associated with risk among individuals with normal weight and overweight, but associations were not statistically significant in the class II and III obesity groups, in which sample sizes were smaller. There were consistently strong associations between IL-6 and all 9 outcomes in both age groups (<65 years and $\geq$ 65 years), as shown in [Supplemental Figure 2](#).

SENSITIVITY ANALYSES. Sensitivity analyses using cohort-specific IL-6 quartiles yielded results consistent with the primary findings ([Supplemental Table 16](#)). Similar results were observed when IL-6 concentrations were standardized within each cohort. The associations also remained robust after excluding the REGARDS cohort ([Supplemental Table 17](#)) and after excluding events occurring within the first 2 years of follow-up to reduce potential reverse causation ([Supplemental Table 18](#)), with effect estimates comparable to those of the main analysis. Finally, analyses using Cox models that included a fixed-effect covariate with robust SEs clustered by cohort produced estimates largely consistent with those from the shared frailty models ([Supplemental Table 19](#)).

DISCUSSION

In this large, diverse, pooled community-based cohort with long-term follow-up, after controlling

TABLE 4 Comparative Associations of IL-6 and hsCRP With Cardiovascular and Mortality Outcomes

	Per SD Increase of log _e (IL-6)	Per SD Increase of log _e (hsCRP)
MI	1.15 (1.10-1.20)	1.08 (1.03-1.13)
Stroke	1.15 (1.10-1.20)	1.08 (1.04-1.13)
AF	1.21 (1.16-1.26)	0.99 (0.95-1.04)
HF	1.23 (1.18-1.28)	1.06 (1.02-1.10)
CVD	1.21 (1.18-1.25)	1.03 (1.00-1.06)
CHD	1.21 (1.17-1.25)	1.04 (1.00-1.08)
All-cause mortality	1.22 (1.20-1.25)	1.03 (1.01-1.05)
CVD-specific mortality	1.31 (1.27-1.35)	1.03 (1.00-1.06)
CHD-specific mortality	1.34 (1.28-1.39)	1.03 (0.99-1.07)

Values are HR (95% CI). Adjusted for sex, age, race/ethnicity, education level, body mass index, prevalent hypertension, antihypertensive medication use, prevalent diabetes, prevalent hyperlipidemia, lipid-lowering medication use, systolic blood pressure, diastolic blood pressure, smoking status, alcohol use, baseline CHD status, per SD increase of log_e(IL-6), and per SD increase of log_e(hsCRP). Baseline CHD status was not adjusted for the incident CHD outcome analysis. Between-cohort variability was addressed by including a cohort-specific frailty term (random effect) in the survival models. Pearson correlation coefficient = 0.441 (95% CI: 0.434-0.448).

Abbreviations as in Tables 1 and 2.

for demographic and clinical covariates, higher circulating concentrations of IL-6 were associated with increased risk for incident cardiovascular events and mortality in a dose-response manner (**Central Illustration**). The harmonized cohort enabled analyses of clinical high-risk subgroups often not feasible in single epidemiologic studies. This revealed that the positive strong associations of IL-6 with cardiovascular outcomes extended to individuals with CKD, prediabetes and diabetes, elevated baseline inflammation, high BMI, and older adults, as well as adults with established CVD. Compared with hsCRP, IL-6 demonstrated stronger associations with incident outcomes. Cardiovascular

risk discrimination improved significantly for most outcomes when IL-6 was added to traditional clinical risk predictors, but the magnitude of improvement was relatively small and warrants cautious interpretation. Results herein derive from analysis of IPD pooled and harmonized over cohorts that span multiple decades and reflect differing eras, treatment patterns, and population health profiles, which is a major strength of our study when compared with others in the extant published literature. Our results strengthen available evidence for IL-6 as a clinically relevant inflammatory biomarker of cardiovascular risk and clearly suggest the potential relevance of emerging IL-6 inhibition therapies for population-level CVD prevention and control.

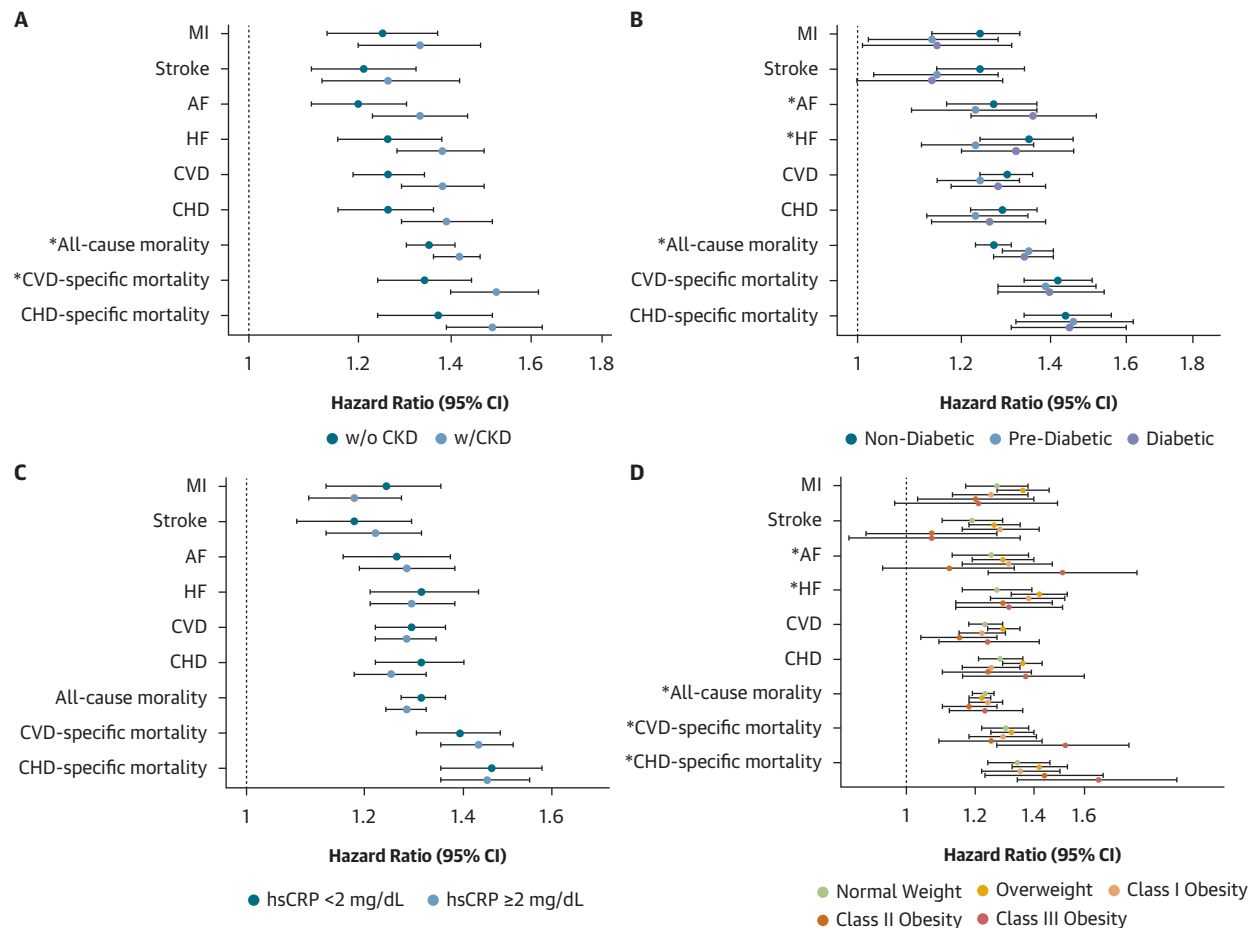
Our findings are generally consistent with the existing literature. Although direct comparisons are constrained by differences in IL-6 assay methods, statistical scaling, and underlying population characteristics, the direction and magnitude of associations observed in our study closely align with those reported in prior meta-analyses and cohort studies. For instance, a 2014 meta-analysis by Kaptoge et al,¹⁸ which included 25 cohorts, revealed a 25% increased risk for nonfatal MI or CHD death per 1-SD increase in IL-6. Other studies have similarly reported especially strong associations between IL-6 and HF as well as cardiovascular mortality,¹⁹⁻²² a pattern we also observed in our analysis. Associations with AF and stroke have been documented in studies such as MESA, CHS, and WHI, which were also included in our pooled analysis.²³⁻²⁶ Our large analysis was completed using pooled, harmonized IPD from 14 well-established, prospective epidemiologic cohorts, which helps overcome several methodological challenges of conventional meta-analyses.¹²⁻¹⁴ Taken together, our results offer a comprehensive

TABLE 5 Discrimination of IL-6 and hsCRP for Cardiovascular and Mortality Outcomes

	Base Model	Base Model + hsCRP	Base Model + IL-6	Base Model + hsCRP + IL-6
MI	0.654 (0.645-0.663)	0.667 (0.657-0.676)	0.665 (0.656-0.674)	0.670 (0.661-0.679)
Stroke	0.672 (0.664-0.681)	0.684 (0.676-0.692)	0.689 (0.681-0.697)	0.692 (0.684-0.700)
AF	0.692 (0.682-0.701)	0.701 (0.691-0.711)	0.692 (0.682-0.702)	0.706 (0.696-0.716)
HF	0.732 (0.724-0.740)	0.750 (0.742-0.758)	0.756 (0.748-0.763)	0.760 (0.752-0.767)
CVD	0.691 (0.683-0.698)	0.700 (0.693-0.707)	0.714 (0.707-0.721)	0.715 (0.708-0.721)
CHD	0.660 (0.651-0.669)	0.671 (0.663-0.680)	0.680 (0.671-0.689)	0.681 (0.673-0.690)
All-cause mortality	0.758 (0.753-0.763)	0.766 (0.761-0.771)	0.775 (0.770-0.780)	0.776 (0.771-0.781)
CVD-specific mortality	0.733 (0.724-0.743)	0.746 (0.737-0.755)	0.768 (0.760-0.778)	0.769 (0.760-0.778)
CHD-specific mortality	0.717 (0.706-0.729)	0.730 (0.719-0.741)	0.752 (0.741-0.762)	0.752 (0.742-0.762)

Values are area under the receiver-operating characteristic curve (95% CI). The base model included sex, age, race/ethnicity, and individual cohort. For each outcome, the comparisons of the area under the curve with the base model resulted in *P* values < 0.05 in all models, with the exception of AF in the base model plus IL-6.

Abbreviations as in Table 2.

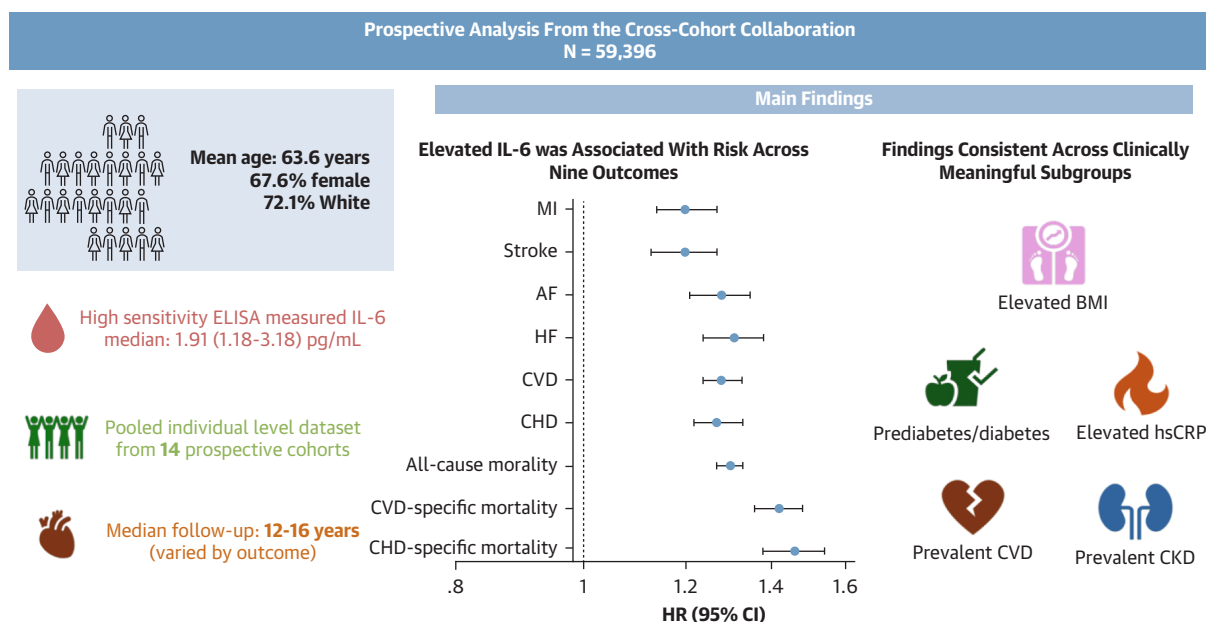
FIGURE 2 Associations of Log-Transformed IL-6 Concentrations With Cardiovascular and Mortality Risks, Stratified by CKD Status, Diabetes Status, hsCRP Level, and BMI Category

(A) Associations by chronic kidney disease (CKD) status. (B) Associations by diabetes status. (C) Associations by high-sensitivity C-reactive protein (hsCRP) level. (D) Associations by body mass index (BMI) category. A, C, and D were adjusted for sex, age, race/ethnicity, education level, BMI, prevalent hypertension, antihypertensive medication use, prevalent diabetes, prevalent dyslipidemia, smoking status, alcohol use, $\log_e(\text{hsCRP})$, and prevalent coronary heart disease. B was adjusted for sex, age, race/ethnicity, education level, BMI, prevalent hypertension, antihypertensive medication use, prevalent dyslipidemia, smoking status, alcohol use, $\log_e(\text{hsCRP})$, and prevalent coronary heart disease. Between-cohort variability was addressed by including a cohort-specific frailty term (random effect) in the models. *Statistically significant interaction term. Abbreviations as in Figure 1.

understanding on the prognostic relevance of IL-6 within the community setting.

The varying strength of IL-6 associations across different cardiovascular outcomes in our study likely reflects several distinct underlying biological pathways. The particularly strong relationship observed with HF aligns with the direct role of IL-6 in promoting myocardial fibrosis and hypertrophy through fibroblast activation and STAT3-mediated transcriptional reprogramming.²⁷ These effects may be especially relevant in the development of HF with preserved ejection fraction, in which inflammation-

driven ventricular stiffening and diastolic dysfunction are key features.²⁸ The comparatively weaker and less consistent results observed for AF in our analyses likely reflect biological differences relative to HF and atherosclerotic outcomes. Our findings for AF are consistent with mechanistic evidence that IL-6 amplifies IL-1 β -induced calcium dysregulation, prolongs atrial calcium transients, and down-regulates connexin-40/43 expression, a process that slows conduction, and promotes arrhythmogenic substrate formation.²⁹ It is important to recognize that part of the association with stroke may be

CENTRAL ILLUSTRATION Pooled Cohorts of Individual-Level Research on IL-6 and Cardiovascular Outcomes and Mortality

Yao Z, et al. JACC. 2026;■(■):■-■.

In a pooled analysis of 14 prospective cohorts (N = 59,396), higher interleukin (IL)-6, measured using high-sensitivity enzyme-linked immunosorbent assay (ELISA), was associated with increased risk for 9 cardiovascular and mortality outcomes over 12 to 16 years of follow-up. Associations were consistent across key cardiometabolic subgroups. AF = atrial fibrillation; BMI = body mass index; CHD = coronary heart disease; CKD = chronic kidney disease; CVD = cardiovascular disease; HF = heart failure; hsCRP = high-sensitivity C-reactive protein; MI = myocardial infarction.

mediated through AF, given that IL-6 also predicts stroke in patients with AF.³⁰ Beyond this arrhythmic pathway, the robust associations we observed for MI and stroke may be explained by the prothrombotic effects of IL-6, including endothelial activation, monocyte recruitment, and hepatic synthesis of fibrinogen and plasminogen activator inhibitor-1.² Importantly, these systemic inflammatory mechanisms operate upstream of CRP, which may explain the incremental predictive value of IL-6 observed in our models beyond the base model of traditional clinical risk predictors and beyond adding hsCRP alone. However, because IL-6 and hsCRP lie along the same inflammatory pathway, the magnitude of improvement in discrimination when considering both markers was statistically significant but relatively small. In addition, IL-6 is also involved in localized inflammatory responses within the atherosclerotic plaque. Cholesterol crystals can trigger the innate immune system by activating the NLRP3 inflammasome, leading to IL-1 β production and subsequent up-regulation of IL-6. This cascade of

direct, plaque-centered inflammation may be particularly relevant in the context of acute plaque rupture, in which impaired fibrinolysis and heightened thrombogenicity can precipitate clinical events from previously subclinical lesions. IL-6's broad systemic effects, including immune dysregulation, catabolic signaling, and metabolic exhaustion, may contribute to multisystem decline and heightened clinical vulnerability beyond the cardiovascular system.³¹ These mechanisms may underlie the strong associations we observed with both cardiovascular and all-cause mortality in our analysis.

Altogether, our findings support IL-6 as a biologically informative and prognostically relevant biomarker that may help refine characterization of population-level cardiovascular risk and inform ongoing research on targeted IL-6 therapeutic strategies. However, they do not imply readiness for routine clinical implementation of IL-6, while there is a gap between mechanistic understanding and current clinical practice. Current clinical guidelines, for example, the 2019 American College of

Cardiology/American Heart Association guideline on the primary prevention of CVD, prioritize hsCRP because of its accessibility and established role as a risk-enhancing factor,³² although it lacks evidence of causal relevance to CVD. In contrast, although with more established biological relevance, IL-6 has not yet been incorporated into clinical guidelines, largely reflecting limitations related to standardization, validation, and real-world implementation. Before broader clinical application becomes feasible, several translational steps are required: harmonization of assays across platforms, establishment and validation of clinically actionable thresholds in broadly representative racial and ethnic groups, and clearer quantification of IL-6's incremental value beyond existing tools. Notably, IL-6 is now measurable using high-throughput, U.S. Food and Drug Administration-cleared immunoassays from multiple vendors, increasing its potential accessibility in clinical practice; however, cross-platform calibration and harmonization remain essential to ensure comparability and consistent interpretation. Until these issues are resolved and prospective evidence becomes available, hsCRP remains the most practical biomarker for inflammatory cardiovascular risk assessment in routine clinical practice.

The major strength of our investigation is the CCC study design itself. Pooling and harmonizing IPD from 14 well-established prospective epidemiologic cohorts afforded the opportunity to evaluate IL-6 in relation to clinical cardiovascular endpoints and mortality on a scale larger than published to date. Because IPD analyses were completed, we minimized the likelihood of several biases that often plague conventional meta-analyses of aggregate data. Standardized measurement of IL-6 using enzyme-linked immunosorbent assays from the same manufacturer across all cohorts strengthens consistency and confidence in our results. We were able to report results in absolute concentrations, rather than the relative units often generated by large proteomic platforms, enhancing interpretability and facilitating direct comparison across studies. The large, diverse pooled cohort offered sufficient sample size and event rates for statistically efficient subgroup analyses. We focused on groups with high cardiometabolic burden and chronically elevated inflammation, in which it is often unclear whether IL-6 simply reflects existing cardiometabolic disease (ie, obesity, diabetes, CKD) or provides additional prognostic information. Our subgroup analyses support the latter, showing that IL-6 was consistently associated with both incident cardiovascular events and mortality across key high-

risk subgroups. Moreover, our results demonstrated that IL-6 remained strongly associated with outcomes among individuals with elevated hsCRP, suggesting that it captures upstream, disease-relevant inflammatory activity not fully reflected by downstream, nonspecific markers. Another notable strength of this study was the inclusion of a broad spectrum of cardiovascular outcomes—including atherosclerotic disease, AF, and HF, which allowed a comprehensive assessment of IL-6 across distinct cardiovascular pathophysiological domains. Our findings therefore consolidate prior cohort evidence and provide a community-level benchmark that may help contextualize results from ongoing and future intervention trials targeting the IL-6 inflammatory pathway.

STUDY LIMITATIONS. Blood IL-6 concentrations were measured at a single time point in most cohorts, which does not allow capture of longitudinal changes in inflammatory status or within-person variability over time. Although the cohorts were composed largely of healthy, community-dwelling participants at average cardiovascular risk, baseline IL-6 levels may reflect underlying disease burden rather than stable long-term inflammatory risk, and reverse causation cannot be fully excluded. As an observational study, our analyses remain subject to potential residual confounding, particularly for exposures such as blood pressure, which may not be fully captured by single measurements, and weight status, which does not differentiate body composition. The net reclassification improvement analyses should be interpreted with caution. The risk cutpoints used for reclassification were chosen to ensure sufficient numbers of events for stable estimation in the present study population and do not reflect established guideline-based clinical thresholds. Consequently, the magnitude of reclassification observed may not directly correspond to clinically actionable changes in risk categorization. Although sensitivity analyses excluding early events in 2 years from baseline yielded consistent results, the possibility of residual reverse causation cannot be entirely excluded, as subclinical CVD present at baseline may influence inflammatory biomarker concentrations years before clinical events occur. Finally, as with all observational analyses, our study cannot establish causality, or directly guide clinical implementation.

Consistent with the prior clinical investigations, our findings further support the relevance of IL-6-mediated inflammation in CVD. In addition to the

results from the RESCUE trial among patients with CKD and elevated hsCRP levels,¹⁰ several large-scale cardiovascular outcome trials are currently under way to assess whether direct inhibition of IL-6 signaling can reduce events in carefully selected high-risk populations, including those with established atherosclerotic disease and CKD, HF with preserved or mildly reduced ejection fraction, and recent MI.³³ In this context, our findings offer community-level insight that complements ongoing interventional efforts, without defining treatment-responsive populations or clinically actionable thresholds. Future trials may also explore IL-6-targeted strategies in other high-risk groups, such as individuals with diabetes or elevated visceral adipose tissue, to better define its role in mitigating residual inflammatory risk across diverse cardiometabolic profiles.

Further research is also needed to determine the cost-effectiveness and optimal use cases for IL-6 measurement. Although universal screening is unlikely to be practical, targeted use in populations with persistently elevated CRP, metabolic dysfunction, or prior cardiovascular events may prove most valuable, particularly if ongoing trials of IL-6 inhibition demonstrate clinical benefit. As the field advances toward precision cardiovascular medicine, IL-6 may serve as a key marker for identifying individuals who are not only clinically high-risk but also biologically predisposed to inflammation-driven vascular events. However, whether IL-6 measurement can meaningfully guide therapeutic decision-making will require confirmation in prospective studies and especially in clinical trials.

CONCLUSIONS

In this large, harmonized analysis of 14 community-based cohorts, higher circulating IL-6 concentrations were consistently associated with a broad range of cardiovascular events and mortality outcomes. These associations were robust across key clinical subgroups as well as in secondary prevention. Compared with hsCRP, IL-6 showed stronger and more consistent associations. IL-6 modestly improved risk discrimination beyond traditional factors and hsCRP. Together, these findings support IL-6 as a key upstream inflammatory biomarker of cardiovascular risk and provide context for ongoing trials targeting IL-6 signaling for CVD prevention.

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APPENDIX For supplemental tables, references, methods, and figures, please see the online version of this paper.